

JetBrains Survey • 23,262 developers

State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

How usage is changing → what developers prefer → what to do about it

Python Closes the Gap: 8 Years of Language Usage Shifts

61% (2024)

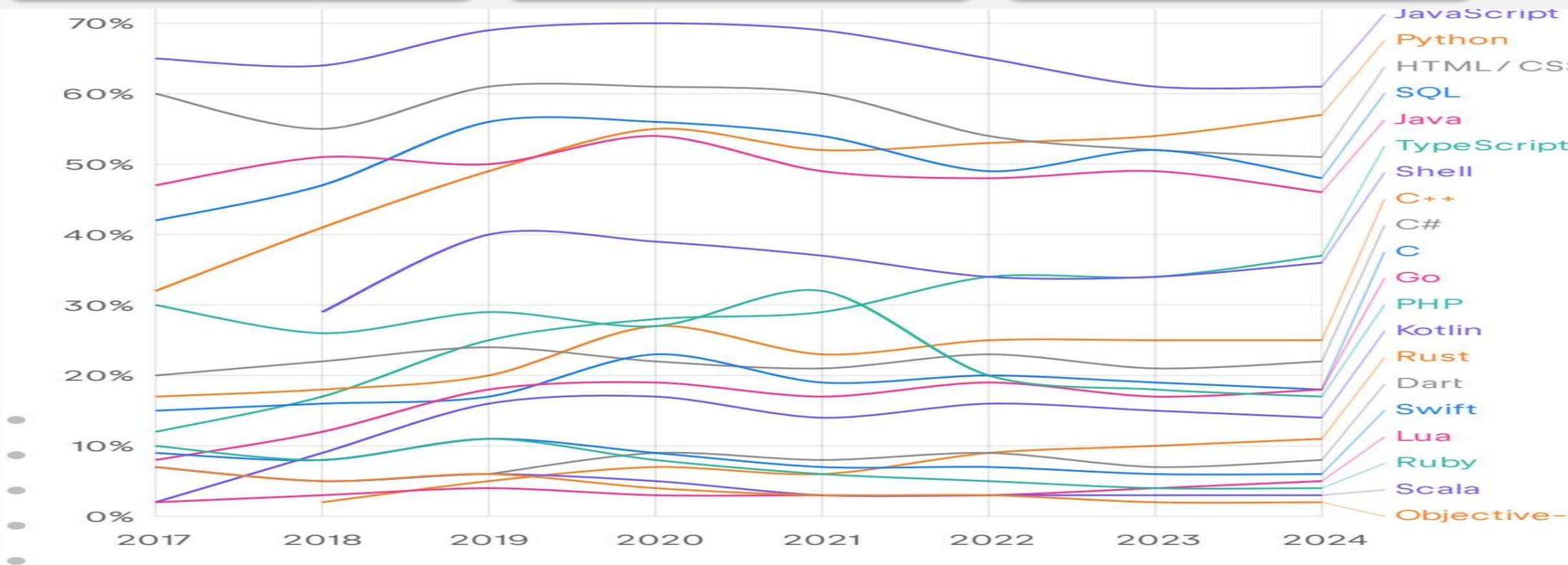
JavaScript usage

57% (2024)

Python usage

37% (2024)

TypeScript usage



Usage trend highlights: JavaScript peaked at 70% (2020) then declined; Python rose from 32% (2017) to 57% (2024); TypeScript grew from 12% (2017) to 37% (2024); PHP fell from 30% (2017) to 17% (2024).

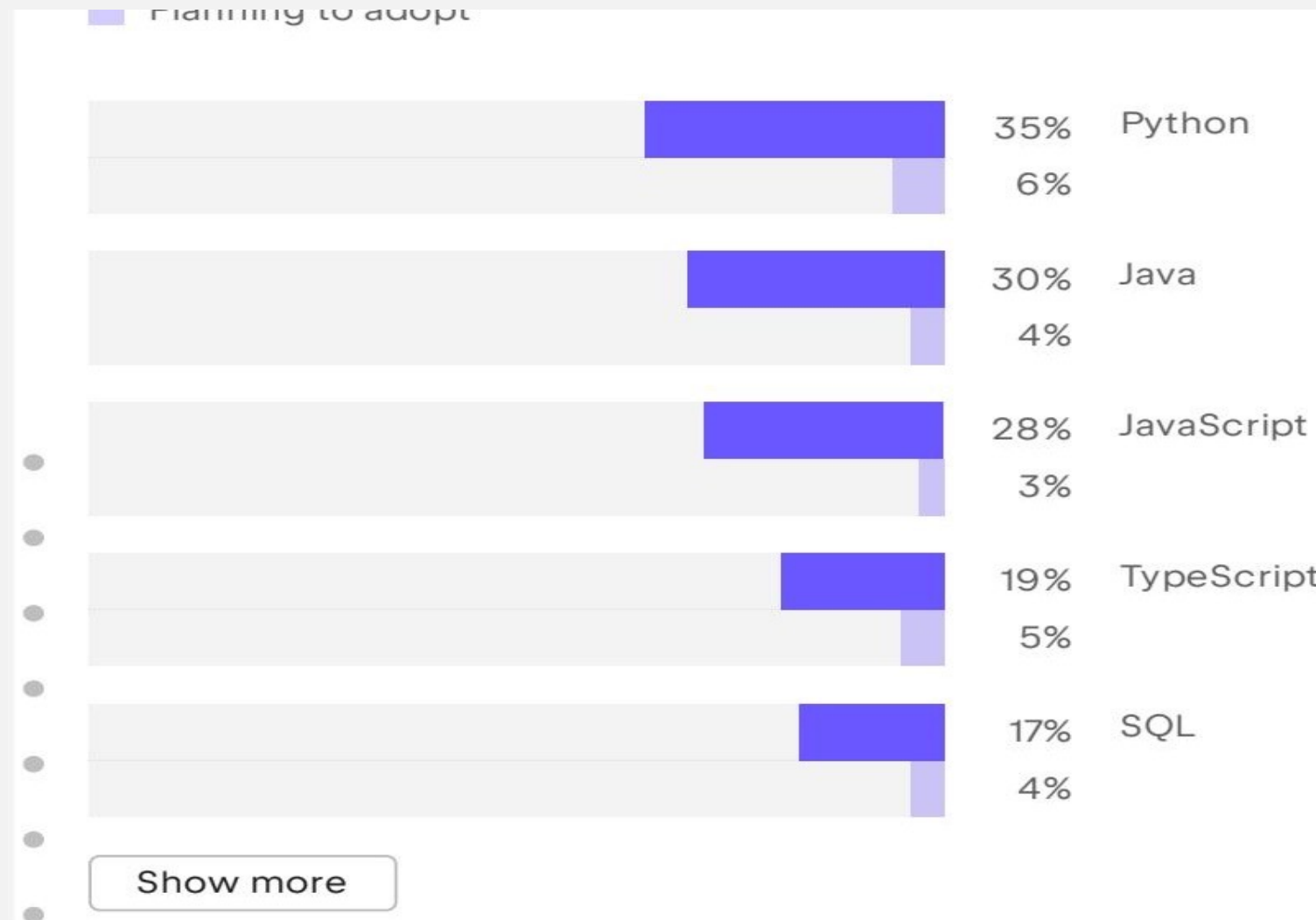
Python Is Now the #1 Primary Language at 35%

Primary stack ≠ planned switch

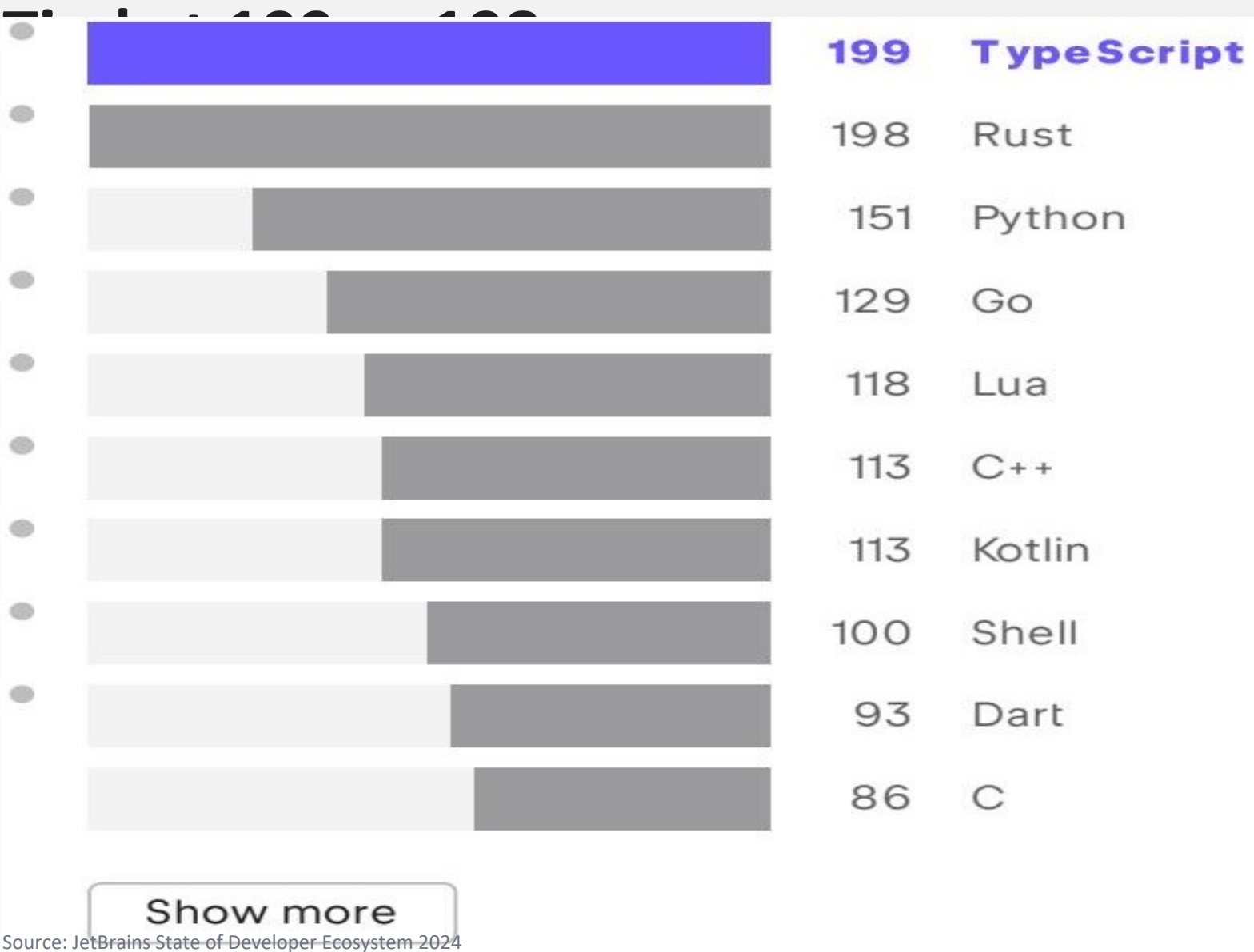
Primary language share (top 5):

- Python 35%
- Java 30%
- JavaScript 28%
- TypeScript 19%
- SQL 17%

Adoption intent stays low (≤6% across languages), implying ecosystem lock-in and a comfort-zone effect.



TypeScript and Rust Top Developer Satisfaction, Nearly



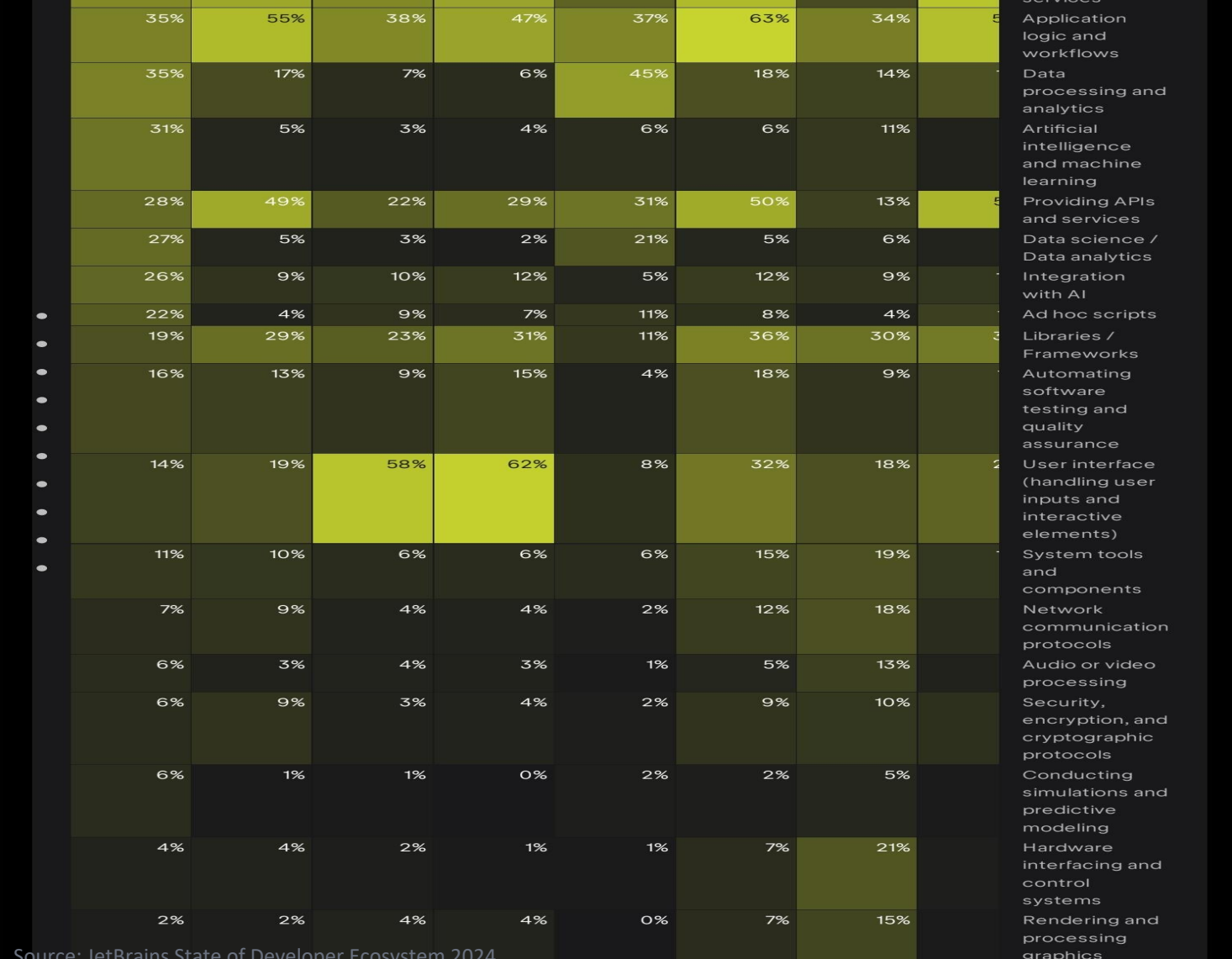
What leaders should infer

Top-10 satisfaction scores:

- TypeScript 199
- Rust 198
- Python 151
- Go 129
- Lua 118
- C++ 113
- Kotlin 113
- Shell 100
- Dart 93
- C 86

High satisfaction clusters around type safety and modern tooling — a retention signal as much as a technical one.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript



Domain leaders (selected)

- AI / Machine Learning → Python (31%)
- Data science / analytics → Python (28%)
- UI development → TypeScript (62%), JavaScript (58%)
- APIs & services → Java (49%)
- Application logic & workflows → Java (55%)
- System tools & components → C++ (27%)

Takeaway: language choice is driven by domain fit more than general preference.

What This Means for Your Technology Strategy in 2025

1) Invest where usage is compounding

Python growth is tied to AI/ML leadership and data use cases.
Staff accordingly: platform + tooling for Python productivity at scale.

2) Standardize the UI stack

TypeScript dominates UI (62%) and leads satisfaction (199).
Treat TS as the default for frontend and full-stack web delivery.

3) Use satisfaction as a retention signal

TypeScript and Rust lead satisfaction (199/198).
Modern tooling + safety correlates with happiness — factor into DX investment.

4) Plan for supply shifts

PHP usage declined from 30% (2017) to 17% (2024).
Optimize legacy cost: migrations, training, and measured hiring expectations.